

Santiago Osorio Jurado

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Profile

Biomedical engineer with expertise in neuroengineering, medical imaging, and machine learning. Extensive experience in R&D, ranging from electrode manufacturing to FDA-compliant clinical trials and deployment of deep learning models to photon-counting CT. Contributed to a pre-print article on brain stimulation technology that can deliver up to 6 mA safely. Skilled in Python, PyTorch, COMSOL, and PCB design. Seeking positions in the medical devices sector in R&D, quality assurance, and software development.

Education

The City College of New York, New York City, NY

B.E. in Biomedical Engineering, Aug 2022 to May 2026

Relevant Coursework: Bioelectrical Circuits, Biomedical Transducers & Instrumentation, Dynamical Systems & Modeling, Image & Signal Processing, Cell & Tissue Mechanics, Transport Phenomena, Experimental Methods, Neural Engineering

Affiliations: Tau Beta Pi, The Engineering Honor Society | Biomedical Engineering Society (BMES)

Professional Experience

Neuroengineering Researcher, CCNY Neural Engineering Group, New York City, NY

Jan 2025 to Present

- Engineered multilayer electrode assemblies for HC-tDCS, incorporating polarity-dependent Zn/AgCl redox layers with hydrogel interface for resistivity of 600 Ohm-cm or lower, allowing for safe currents of up to 6 mA, three times higher than the standard 2 mA dosages.
- Conducted characterization studies of the electrodes through EIS and SEM analysis at various dosages from 1 to 10 mA, confirming electrochemical stability and overpotential characteristics for therapeutic applications.
- Performed IRB-compliant human subject experiments involving 5 healthy volunteers through single-blind, randomized trials from 1 to 6 mA in 30-minute sessions, gathering safety measurements including VAS pain ratings, Draize erythema assessment, thermal imaging, and PANAS testing.
- Modeled electric fields within the cortex of 1.95 to 3.25 V/m at 6 mA using COMSOL FEM simulations based on MRI head geometry; analyzed dose-response relationships using linear mixed-effects models coded in Python.
- Engineered custom PCB boards for adaptive hybrid voltage and current-controlled stimulator hardware; published bioRxiv preprint manuscript (doi.org/10.1101/2025.06.11.659142) demonstrating tolerability at maximum doses of 6 mA.

Machine Learning Researcher, UC Berkeley Computational Imaging Lab, Berkeley, CA

Jun 2025 to Dec 2025

- Trained both U-Net and CNN-transformer model hybrids using PyTorch on 768 image pairs to improve the quality of extreme short exposure darkfield microscopy images for biological applications.
- Developed a synthetic dataset generation pipeline by implementing gamma correction and simulating Poisson and Gaussian noise in order to create realistic data for label-free live cell imaging purposes.
- Compared performance metrics including PSNR, SSIM, and error maps between U-Net and CNN-transformer models and found that the former was more effective in noise removal.
- Implemented both deep learning models in a system running embedded Linux, providing end-to-end training and deployment code.

Medical Imaging Researcher, Royal Institute of Technology, Stockholm, Sweden

May 2024 to Sep 2024

- Evaluated the impact of tube voltage (70-140 kVp) on the image quality of Photon-Counting CT scans using Gammex multi-energy phantoms containing iodine at concentrations of 0.5, 2, and 20 mg/ml using a Siemens Naeotom Alpha CT system.
- Found that 90 kVp and 40 keV monoenergetic virtual reconstruction were the ideal conditions for iodine contrast-to-noise ratio at two different levels of radiation doses, 5.98 mGy and 12.1 mGy CTDIvol.
- Created an automated Python workflow for comparing image quality measures for the different energy levels and reconstruction modes (QIR0/QIR4).
- Developed a clinical decision guide based on physics optimizations for helping radiologists select CT scan parameters.

Biology Laboratory Assistant, The City College of New York, New York City, NY

Aug 2023 to May 2024

- Helped more than 100 undergraduates per semester by tuning up laboratory equipment such as microscopes, pipettes, incubators, and spectrophotometers.
- Prepared reagents, biological and chemical, in adherence to BSL-2 safety guidelines and maintained a perfect record on that for two consecutive semesters.
- Maintained chemical inventories, ensuring that all regulations were adhered to and no expiration or loss occurred.

Skillsets

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| • Programming: Python, MATLAB, PyTorch, TensorFlow, SQL | • Modeling & Simulation: COMSOL Multiphysics, FEM, U-Net, deep learning pipelines | • Medical Device: IRB protocols, design controls, clinical data collection | • Hardware & Prototyping: PCB design, Arduino, CAD (Fusion 360), circuit design |
| • Laboratory: EIS, SEM, BSL-2, sterile technique, instrument calibration | | | |